



Artificial intelligence adoption and corporate financial risk

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ABSTRACT

Artificial intelligence (AI) is a critical driver of competitive advantage. Using data from China's A-share listed companies (2010–2022), this study finds that AI adoption significantly reduces corporate financial risk through structural, resource-based, and technological channels—especially in nonhigh-tech firms and industries with lower uncertainty.

1. Introduction

Rapid digital advancement and complexity present firms with both challenges and opportunities. Hence, companies must innovate to improve quality, reduce costs, increase efficiency, and strengthen user engagement (Igna and Venturini, 2023; Babina et al., 2024; Zhang et al., 2025). In this context, artificial intelligence (AI) is a transformative force in building value propositions and unlocking opportunities (Mustak et al., 2021; Czarnitzki et al., 2023; Zhao et al., 2025). Corporate AI adoption rose from 20 % in 2017 to 50 % in 2022, with most firms allocating >5 % of their budgets to AI initiatives. Moreover, about 23 % applied at least 5 % of their 2022 earnings before interest and taxes AI.

The literature documents various business impacts of AI adoption. It lowers production costs, boosts efficiency, and enables cost-leading firms to earn excess profits (Bosse et al., 2023; Wu et al., 2025). AI automates traditional production inputs, reduces labor needs and defect rates, and improves product quality. Integrated with technologies such as the Internet of Things and cloud computing, AI supports intelligent manufacturing (Li et al., 2023), shortens development cycles, and raises innovation efficiency (Bahoo et al., 2023). However, AI's potential for mitigating corporate risk remains underexplored.

Traditional risk models, dependent on historical data and human judgment, often fail to respond to dynamic markets and are prone to error (Kemp, 2024). AI enables real-time monitoring of financial health, better risk forecasting, and smarter decision-making, all of which reduce financial risk. Tools like robotic process automation and intelligent data analysis further limit operational mistakes and improve efficiency, bolstering resilience. Assessing AI's influence on corporate financial risk is therefore critical. This study analyzes data from listed firms between 2010 and 2022 to evaluate AI's impact on financial risk amid a complex economy. Findings show AI enhances corporate risk resistance through structural, resource, and technological channels, especially in nonhigh-tech sectors and industries with lower uncertainty.

Although prior research has focused on AI's role in intelligent manufacturing and human resource (HR), few have explored its function in risk resilience. We investigate AI's importance in this area, demonstrating its benefits under specific conditions, and present empirical evidence on how it reinforces financial risk management. Second, we examine these effects in complex economic conditions, providing a theoretical framework explaining how emerging technologies transform risk management.

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